DOCKHUB · FEASIBILITY ASSESSMENT

DockHub feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First feasibility assessment at the kit 1.9.1 upgrade, for the current design (option A) at TRL 3

DockHub feasibility assessment

Closed at TRL 3: feasible on paper, with three risks that only hardware can retire. The calculation note (DKH-CAL-001) shows 11 of 16 original requirements met, a 31 s swap, 1.25 kW peak grid draw and a station that stays upright in a 30 m/s gust. The cabinet is built from folded sheet steel, off-the-shelf certified chargers and hobby-grade controllers, and the model (cad/src/model.py, D3) has 537 parts that fit with no interference. What remains unproven is fire containment (PR-008), charging on hot days (PR-009) and cost: the parts come to USD 1,519 against the USD 1,200 target. Method: STANDARDS section 19.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Four bays in one riveted cabinet, a compartment, cradle, receptacle, charger and CAN channel each; 31 s swap; 36 swaps a day ceiling; 234.5 kg for the two-bay prototype in the model	High	Two bays serve only 9 of the 20 daily swaps; the four-bay fit is needed for the duty case	DKH-STM-001 S1 to S8; DKH-CAL-001 s2 to s6
Materials	1.5 mm galvanized steel sheet, stainless hinges and anchors, EPDM gaskets, a printable flame-retardant cradle	High	Cradle filament rated UL 94 V-0 to confirm; coating life in road salt (S19)	DKH-STM-001 S19; DKH-DEC-001 to-confirm item 8
Manufacturing and assembly	Laser-cut and folded panels, blind rivets, a welded channel plinth, bolted canopy; build plan with 12 stages	Medium	Needs a sheet metal shop and a welder; tolerances at the folds to confirm	DKH-BLD-001 (docs/05-build-plan.md); cad/src/model.py
Power and controls	One 120 V 15 A or 230 V 10 A circuit; four chargers with power factor 0.9 or better; a controller with a CAN channel per bay	Medium	At 120 V the draw is 11.0 A with PFC chargers, 17.2 A without: the chargers must have PFC	DKH-CAL-001 s6; DKH-STM-001 S3
Safety	Detector, condensed aerosol, steel compartments, vent plenum and hood	Low	Runaway propagation cannot be excluded on paper (S11); aerosol retention (S13); fumes at the door (S23)	DKH-STM-001 S11, S13, S23; DKH-RSK-001
Supply of parts	Certified chargers, a controller, a 400 W panel and the SwapCell pack are bought; sheet and plinth from a local shop	Medium	The pack is the SwapCell project's (USD 414 each, not in the cost); charger PFC and footprint to confirm	bom/bom.csv; DKH-DEC-001 to-confirm items 1 to 9
Cost	USD 1,519 with anchors and pad, USD 1,214 without, USD 1,743 with all four bays	Low	Over the target by USD 319 (PR-016); the hypothetical control target is Amish's to keep or move	bom/bom.csv; DKH-CAL-001 s12
Site and permits	Private forecourt with a single-phase circuit; a pad and anchors; fire review	Medium	A public sidewalk needs consent, a Certificate of Approval route that treats a swap cabinet as a charging cabinet is unconfirmed	DKH-REG-001 G1, G4, G5
Market	Open, owner-run swap cabinets for a delivery workforce that has few safe places to charge; operators exist	Medium	A closed incumbent network may crowd out an open one; pack supply depends on SwapCell	DKH-MKT-001

Ratings: High, Medium, Low.